

SOLUTION

Depth First Search (DFS) explores a graph by moving as deep as possible along each branch before backtracking. It begins at a specified starting vertex and marks it as visited.

From the current vertex, the algorithm selects an unvisited adjacent vertex and continues the traversal recursively (or using a stack). If a vertex has no unvisited neighbors, the algorithm backtracks to the previous vertex and continues exploring the remaining unexplored paths.

This process continues until all vertices reachable from the starting vertex have been visited. To avoid revisiting the same vertex and entering infinite loops in cyclic graphs, a **visited** list or set is maintained throughout the traversal.

DFS is suitable for solving problems that require exploring every possible path, searching for a solution in large state spaces, detecting cycles, finding connected components, generating spanning trees, and performing topological sorting.

The algorithm can be implemented using either recursion or an explicit stack, both of which follow the Last-In-First-Out (LIFO) principle.